

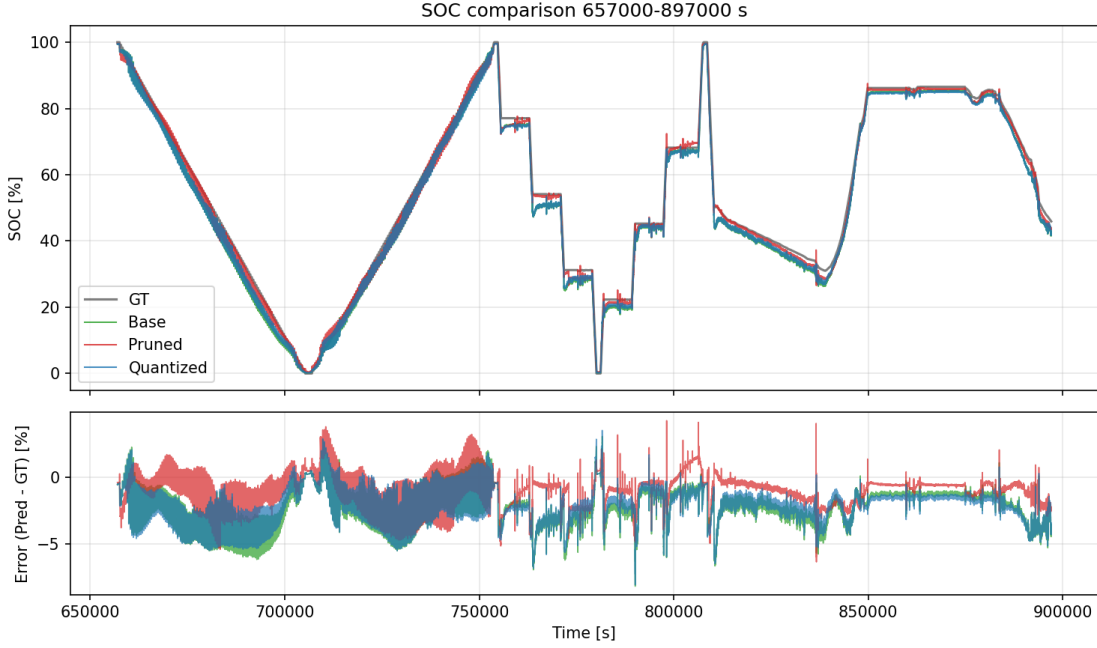


Figure 7.11: Detailed check-up-region analysis of embedded SOC streaming replay. The figure covers a representative diagnostic interval with long integration horizons and illustrates the stability of the stateful deployments during extended charge/discharge behaviour.

7.10 Software-Level Fault-Sensitivity Results

Holding the last valid estimate during randomly unavailable output reports has almost no effect on aggregate accuracy for report-loss rates up to 10%. In every model, the resulting MAE change remains below 0.001 percentage points. Longer deterministic report gaps expose a clearer SOC effect. Ten gaps of 3600 samples each increase SOC MAE by approximately 0.05 percentage points, whereas the slowly changing SOH estimate remains almost unaffected. Since inference continues beneath the reporting layer, this part of the test characterises output availability and not corruption of the recurrent calculation.

The one-sample input-buffer experiment produces a distinct transient response. Across the SOC variants, median peak deviations between the faulted and clean branches lie between 3.29 and 3.82 percentage points. Their P95 values range from 16.18 to 23.81 percentage points. Every one of the 30 events per model satisfies the recovery criterion within 60 s, with median recovery times between 10.0 and 12.5 s. Quantized behaves similarly to Base, while Pruned combines the smallest median